4.7.2 Electricity

A quantitative study of electric currents in circuits leads on to an introduction of the alternating currents used in the mains electricity supply. Electricity is distributed to consumers via the National Grid. The rate of energy transfer depends on the power of the appliances connected to the supply.

In this topic students have to be able to apply the equations relating potential difference, current, quantity of charge, resistance, power, energy and time, and solve problems for circuits which include resistors in series, using the concept of equivalent resistance (MS 1c, 3b, 3c, 3d).

There are two required practicals in this topic. One is to investigate the I–V characteristics of a variety of circuit elements at constant temperature. The other is to set up and use appropriate circuits to investigate factors that affect the resistance of an electrical component. The practical activities give students experience of the apparatus and techniques related to the use of circuit diagrams to construct and check series and parallel circuits that include a variety of common circuit components.

4.7.2.1 Electric current

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that current is a rate of flow of charge, that for a charge to flow a source of potential difference and a closed circuit are needed and that a current has the same value at any point in a single closed loop. Recall and use the relationship between quantity of charge, current and time.	For electrical charge to flow through a closed circuit the circuit must include a source of potential difference. Electric current is a flow of electrical charge. The size of the electric current is the rate of flow of electrical charge. $charge\ flow = current \times time$ [$Q = I\,t$] charge flow, Q, in coulombs, C current, I, in amperes, A (amp is acceptable for ampere) time, t, in seconds, s A current has the same value at any point in a single closed loop.	WS 3.3, MS 3b, 3c Recall and apply this equation.

4.7.2.2 Current, resistance and potential difference

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that current (I) depends on both resistance (R) and potential difference (V) and the units in which these are measured; recall and apply the relationship between I, R and V, and explain that for some resistors the value of R remains constant.	The current through a component depends on both the resistance of the component and the potential difference across the component. The greater the resistance of the component the smaller the current for a given potential difference across the component. $potential\ difference = current \times resistance$ $[\ V = I\ R\ ]$ potential difference, *V*, in volts, V current, *I*, in amperes, A (amp is acceptable for ampere) resistance, *R*, in ohms, Ω The current through an ohmic conductor (at a constant temperature) is directly proportional to the potential difference across the conductor. This means that the resistance remains constant as the current changes.	WS 3.3, MS 3c Recall and apply this equation.
Explain that in other types of resistor the value of R can change as the current changes; explain the design and use of circuits to explore such effects – including lamps, diodes, thermistors and light-dependent resistors (LDRs).	The resistance of components such as lamps, diodes, thermistors and LDRs is not constant; it changes with the current through the component. The resistance of a filament lamp increases as the temperature of the filament increases. The current through a diode flows in one direction only. The diode has a very high resistance in the reverse direction. The resistance of a thermistor decreases as the temperature increases. The resistance of an LDR decreases as light intensity increases.	WS 1.2, 3.5, MS 4c, 4d, 4e Use graphs to determine whether circuit components are linear or non-linear and relate the curves produced to the function and properties of the component.

Required practical activity 15: use circuit diagrams to construct appropriate circuits to investigate the I–V characteristics of a variety of circuit elements including a filament lamp, a diode and a resistor at constant temperature.

AT skills covered by this practical activity: physics AT 6 and 7.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in Key opportunities for skills development (page 172).

Required practical activity 16: use circuit diagrams to set up an appropriate circuit to investigate the factors affecting the resistance of an electrical component. This should include:

- the length of a wire at constant temperature
- combinations of resistors in series and in parallel.

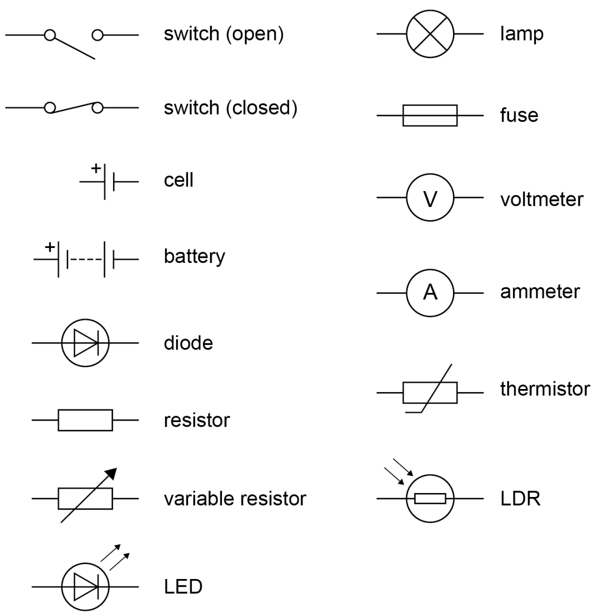
AT skills covered by this practical activity: physics AT 1, 6 and 7.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in Key opportunities for skills development (page 173).

4.7.2.3 Series and parallel circuits

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the difference between series and parallel circuits; explain why, if two resistors are in series, the net resistance is increased, whereas with two in parallel the net resistance is decreased (qualitative explanation only). Calculate the currents, potential differences and resistances in direct current (dc) series circuits, and explain the design and use of such circuits for measurement and testing purposes.	There are two ways of joining electrical components: in series and in parallel. Some circuits include both series and parallel parts. For components connected in series: • there is the same current through each component • the total potential difference of the power supply is shared between the components • the total resistance of two components is the sum of the resistance of each component. $R_{total} = R_1 + R_2$ resistance, R, in ohms, Ω For components connected in parallel: • the potential difference across each component is the same • the total current through the whole circuit is the sum of the currents through the separate components • the total resistance of two resistors is less than the resistance of the smallest individual resistor.	WS 3.3, MS 1c, 3b, 3c, 3d Solve problems for circuits which include resistors in series using the concept of equivalent resistance. WS 3.3, MS 1c, 3b, 3c, 3d Calculate the currents, potential differences and resistances in dc series circuits. Calculating the total resistance of two resistors joined in parallel is not required.

4.7.2.4 Circuit elements

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Represent in dc series circuits with the conventions of positive and negative terminals, the symbols that represent common circuit elements, including diodes, LDRs and thermistors.	Circuit diagrams use standard symbols.  	

4.7.2.5 Direct and alternating currents

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that the domestic supply in the UK is ac, at 50Hz and about 230 volts; explain the difference between direct and alternating voltage.	Cells and batteries supply current that always passes in the same direction. This is called direct current (dc). An alternating current (ac) is one that changes direction. In the UK ac supply the current changes direction 50 times per second.	

4.7.2.6 Mains cables

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall the differences in function between the live, neutral and earth mains wires, and the potential differences between these wires; hence explain that a live wire may be dangerous even when a switch in a mains circuit is open, and explain the dangers of providing any connection between the live wire and earth.	Most electrical appliances are connected to the mains using three-core cable. The insulation covering each wire is colour coded for easy identification. • Live wire – brown • Neutral wire – blue • Earth wire – green and yellow stripes The live wire carries the alternating potential difference from the supply. The neutral wire completes the circuit. The earth wire is a safety wire to stop the appliance becoming live. The potential difference between the live wire and earth (0 V) is about 230 V. The neutral wire is at or close to earth potential (0 V). The earth wire is at 0 V; it carries a current only if there is a fault. Our bodies are at earth potential (0 V). Touching the live wire produces a large potential difference across our body. This causes a current to flow through our body, resulting in an electric shock.	WS 1.5 Identify an electrical hazard in a given context.

4.7.2.7 Power

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain, with reference to examples, the definition of power as the rate at which energy is transferred.	Power is defined as the rate at which energy is transferred or the rate at which work is done. $power = \frac{energy\ transferred}{time}$ $\left[P = \frac{E}{t}\right]$ $power = \frac{work\ done}{time}$ $\left[P = \frac{W}{t}\right]$ power, *P*, in watts, W energy transferred, *E*, in joules, J time, *t*, in seconds, s work done, *W*, in joules, J An energy transfer of 1 joule per second is equal to a power of 1 watt.	WS 1.2, 3.3,MS 3b, 3c Recall and apply both of these equations.
Explain how the power transfer in any circuit device is related to the potential difference across it and the current, and to the energy changes over a given time.	The power of an electrical device is related to the potential difference across it and the current through it by the equation: $power = potential\ difference \times current$ $[P = V\ I]$ $power = (\text{current})^2 \times resistance$ $\left[P = I^2\ R\right]$ power, *P*, in watts, W potential difference, *V*, in volts, V current, *I*, in amperes, A (amp is acceptable for ampere) resistance, *R*, in ohms, Ω	WS 1.2, 3.3, MS 3b, 3c Recall and apply both of these equations.

4.7.2.8 Power and domestic electric appliances

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe how, in different domestic devices, energy is transferred from batteries and the ac mains to the energy of motors or of heating devices. Describe all the changes involved in the way energy is stored when a system changes, for common situations: bringing water to a boil in an electric kettle. Describe, with examples, the relationship between the power ratings for domestic electrical appliances and the changes in stored energy when they are in use. Describe and calculate the changes in energy involved when a system is changed by work done when a current flows.	Everyday electrical appliances are designed to bring about energy transfers. The amount of energy an appliance transfers depends on how long the appliance is switched on for and the power of the appliance. Work is done when charge flows in a circuit. The amount of energy transferred can be calculated using the equations: $energy\ transferred = power \times time$ $[\ E = P\ t\ ]$ $energy\ transferred = charge\ flow \times potential\ difference$ $[\ E = Q\ V\ ]$ energy transferred, *E*, in joules, J power, *P*, in watts, W time, *t*, in seconds, s charge flow, *Q*, in coulombs, C potential difference, *V*, in volts, V	WS 1.4 Explain everyday and technological applications of science. WS 1.2, 3.3, MS 3c Recall and apply both of these equations.

4.7.2.9 The National Grid

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that, in the national grid, electrical power is transferred at high voltages from power stations and then transferred at lower voltages in each locality for domestic use; and explain how this system is an efficient way to transfer energy.	The National Grid is a system of cables and transformers linking power stations to consumers. Electrical power is transferred from power stations to consumers using the National Grid. Step-up transformers are used to increase the potential difference from the power station to the transmission cables then step-down transformers are used to decrease the potential difference to a much lower value for domestic use. (HT only) Students should be able to select and use the equation: potential difference across primary coil x current in primary coil = potential difference across secondary coil x current in secondary coil as given on the equation sheet.	WS 1.4 Explain everyday and technological applications of science. Detailed knowledge of the structure of a transformer is not required.